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(1)

Measures of Variations (Dispersions)

Range, Inter Quartile

Range

& Quartile Deviation

5.1 Definition and purposes

In a series of data, all the items or observations are not equal. There is difference or variation among the values. The degree of variation is evaluated by various relative and absolute measures of dispersion. Small dispersion indicates high uniformity of the items, while large dispersion indicates less uniformity.

Definition and purposes (Contd. ...)

Definition 1:- Dispersion (or variation) is the extent to which values in data differ among themselves.

OR Variability provides a quantitative measure of the degree to which scores in the distribution are **spread out or clustered together.**

Definition and purposes (Contd. ...)

Example 1:- Consider the following observations and find the mean, the median and the mode.

i) 6, 9, 3, 17, 10

ii) 2, 14, 15, 5, 9

iii) 7, 12, 1, 9, 16

The easiest way to differentiate relative dispersion/absolute dispersion is to check whether your statistic involves **units**.

Absolute measures always have units, while *relative measures do not*.

Absolute Measures of Dispersion

Absolute measures of dispersion use the original units of data and are most useful for understanding the dispersion within the context of your experiment and measurements.

Relative Measures of Dispersion

Relative measures of dispersion, are also known as *coefficients of dispersion* are obtained as *ratios or percentages*. These are pure numbers *independent of units of measurement* and used to compare two or more sets of data values.

Relative Measures of Dispersion (Contd ...)

For example, when rainfall data is made available for different days in *mm*, any absolute measures of dispersion give the variation in rainfall in *mm*. On the other hand, *relative measures of dispersion are free from the units of the measurements of the observations.*

Absolute Measures of Dispersion

Absolute measures of variation include:

- Range
- Inter Quartile Range
- Quartile Deviation
- Average Deviations
- Standard Deviation
- Variance

Relative Measures of Dispersion

And ***relative measures of variation include:***

- Coefficient of Range
- Coefficient of Quartile Deviation
- Coefficient of Average Deviations
- Coefficient of Variation

Positional Measures of Variations

i) *The Range and Coefficient of Range*

Definition: The range is defined to be difference between the largest a smallest value.

$$\text{i.e. } R = x_{\max} - x_{\min}$$

Positional Measures of Variations

Example2:- Consider example 1 and find the range for each of them.

i) 6, 9, 3, 17, 10 $\Rightarrow R = 17 - 3 = 14$

ii) 2, 14, 15, 5, 9 $\Rightarrow R = 15 - 2 = 13$

iii) 7, 12, 1, 9, 16 $\Rightarrow R = 16 - 1 = 15$

Note that the range is the *simplest and quick* reference of variability.

Positional Measures of Variations

Example 3: The following data refer to the test score (out of 10) of two different sections. Find the range and discuss on your result.

i) 1, 10, 9, 10, 8, 9

ii) 8, 10, 1, 2, 6, 4

Example 4:- Find the range for the following data and discuss on your result.

6, 2, 3, 3, 6, 2, 2, 209, 6, 3, 5, 6

Positional Measures of Variations

Coefficient of Range

$$\text{Coefficient of Range} = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$$

Example 5:- Consider example 2 and find the coefficient of range for each of them.

Positional Measures of Variations

$$\text{i) } 6, 9, 3, 17, 10 \quad \Rightarrow \quad R = 17 - 3 = 14$$

$$\text{ii) } 2, 14, 15, 5, 9 \quad \Rightarrow \quad R = 15 - 2 = 13$$

$$\text{iii) } 7, 12, 1, 9, 16 \quad \Rightarrow \quad R = 16 - 1 = 15$$

Positional Measures of Variations

$$\text{Coefficient of Range (i)} = \frac{17 - 3}{17 + 3} = \frac{14}{20} = 0.7$$

$$\text{Coefficient of Range (ii)} = \frac{15 - 2}{15 + 2} = \frac{13}{17} = 0.76$$

$$\text{Coefficient of Range (iii)} = \frac{16 - 1}{16 + 1} = \frac{15}{17} = 0.88$$

Positional Measures of Variations

Merits and Demerits of Range

• *Merits or Uses:*

1. It is easy to calculate and simplest to understand even for a beginner.
2. It is one of those measures which are rigidity defined.

Positional Measures of Variations

3. It gives us the total picture of the problem even with a single glance.
4. Meteorological Department also makes forecast about the weather by keeping range of temperature, in view.

Positional Measures of Variations

• *Demerits or Limitations :*

1. Range is not based on all the terms. Only extreme items reflect its size. Hence range cannot be completely representative of the data as all other middle values are ignored.
2. Due to the above reason range is known as a *crude and unreliable* measure of dispersion.

Positional Measures of Variations

ii) The Interquartile Range (IQR)

Definition:- The Interquartile range (IQR) is defined to be the difference of the upper and lower quartiles.

$$\text{i.e. IQR} = Q_3 - Q_1$$

- *The IQR is mostly used when there are extreme values in the Data.*

Positional Measures of Variations

For the same purpose we also can use the IP which is defined by

$$\text{IPR} = P_{90} - P_{10}$$

But since IQR avoids 50% of the data (25% to the right and 25% to the left), it is not capable of further algebraic treatment.

Positional Measures of Variations

ii) The Quartile Deviation and Coefficient of Quartile Deviation.

Definition: The Quartile deviation, denoted by QD is defined to be half of the IQR.

$$\text{i.e. QD} = \frac{IQR}{2} = \frac{Q_3 - Q_1}{2}$$

Positional Measures of Variations

Note that about 50% of the observations are approximately between $\bar{x} - \text{QD}$ and $\bar{x} + \text{QD}$.

Example 6:- Consider the data given in chapter 3 and check the above statement.

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Positional Measures of Variations

Example 7:- Consider the data given in chapter 3 again and find the coefficient of QD.

Merits and Demerits of QD

Merits or uses:

- It can be easily calculated and simply understood.
- It does not involve much mathematical difficulties.
- As it takes middle 50% terms hence it is a measure better than Range and Percentile Range.

Positional Measures of Variations

- It is not affected by extreme terms as 25% of upper and 25% of lower terms are left out.
- Quartile Deviation also provides a short cut method to calculate Standard Deviation using the formula

$$6 \text{ Q.D.} \approx 5 \text{ M.D.} \approx 4 \text{ S.D.}$$

Positional Measures of Variations

Demerits or Limitations:

- As Q_1 and Q_3 are both positional measures hence are not capable of further algebraic treatment.
- It is too much affected by fluctuations of samples.

Positional Measures of Variations

- 50% terms play no role; first and last 25% items ignored may not give reliable result.
- We can't call it a measure of dispersion as it does not show the scatterness around any average.

Mathematical Measures of Variations

Average Deviations

In order to understand what the average deviation is, you should understand what the term **absolute deviation** is. Absolute deviation is the distance between each value in the data set and that data set's *mean or median*. To find the distance you have to subtract the values.

Mathematical Measures of Variations

For example, suppose you have 5 values:

61, 52, 55, 58, 54 where $\bar{x} = 56$. The deviations (

$x_i - \bar{x}$) are:

$$61 - 56 = 5$$

$$52 - 56 = -4$$

$$55 - 56 = -1$$

Mathematical Measures of Variations

The sum of these deviations is zero; this must always be so in any set of data since it follows from the property of $\bar{x}$ that

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Mathematical Measures of Variations

We shall therefore use the absolute value (i.e. the value ignoring the sign) of each deviation to form an average. The average deviation of the given data set is

$$\frac{|5| + |-4| + |-1| + |2| + |-2|}{5} =$$
$$\frac{5 + 4 + 1 + 2 + 2}{5} = 2.8$$

Mathematical Measures of Variations

Definition 1:- The average deviation (about the mean) of a set of N observations $x_1, x_2, x_3, \dots, x_N$ is the average of the absolute deviation from the mean and equals

$$\text{Average deviation(about the mean)} = \frac{\sum_{i=1}^N |x_i - \bar{x}|}{N}$$

Mathematical Measures of Variations

An alternative definition of average deviations uses deviation *about the median* instead of the mean as

$$\text{Average deviation(about the median)} = \frac{\sum_{i=1}^n |x_i - \tilde{x}|}{N}$$

Mathematical Measures of Variations

Note that we use this deviation if the data have extreme values (since the mean is affected by extreme values).

Example1:- Find the average deviation about the mean and the median for the following data set:

14, 6, 5, 9, 1, 4, 3, 9, 12

Mathematical Measures of Variations

Coefficient of Average Deviations

$$\text{Coefficient of MD}(\bar{x}) = \frac{\text{MD}(\bar{x})}{\bar{x}}$$

$$\text{Coefficient of MD}(\tilde{x}) = \frac{\text{MD}(\tilde{x})}{\tilde{x}}$$

Example2:- Find the coefficient of $\text{MD}(\bar{x})$ and coefficient of $\text{MD}(\tilde{x})$ for the above data set given in example 1.

Mathematical Measures of Variations

Merits:

- It is simple to understand and easy to compute.
- It is based on each and every item of the data.
- MD is less affected by the values of extreme items than the Standard deviation.

Mathematical Measures of Variations

Demerits:

- The greatest drawback of this method is that algebraic signs are ignored while taking the deviations of the items.
- It is not capable of further algebraic treatments.
- It is much less popular as compared to standard deviation.

Any Questions



Any Questions

